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Gothenburg, SWEDEN

PROJECTS

AquaSvea Simulation Tool

University of Gothenburg AquaSvea AB

Gothenburg, SWEDEN

- Developed an interactive desktop application for simulation of wastewater treatment processes in collaboration with AquaSvea AB.
- Designed and implemented a modular simulation architecture in Java using object-oriented design principles to separate input handling, calculation logic, validation, and result presentation.
- Transformed Excel-based engineering models and domain-specific process calculations into a structured and extensible software system.
- Built a multi-step JavaFX wizard interface allowing users to configure different wastewater treatment flows and process parameters interactively.
- Implemented validation systems with dynamic visual feedback to highlight acceptable and non-acceptable simulation values based on engineering constraints.
- Developed a YAML-driven configuration system that enables domain experts to define, modify, and extend simulation calculations without changing application source code.
- Implemented dynamic loading and execution of user-defined calculation models, improving maintainability and reducing developer dependency for future process updates.
- Developed PDF report export functionality using Thymeleaf and OpenHTMLToPDF for generation of simulation summaries and technical documentation.
- Added JSON-based persistence for saving and loading simulation runs, enabling reproducibility, comparison of process configurations, and local storage of simulation history.
- Worked iteratively using Agile-inspired sprint planning, Trello task management, stakeholder feedback sessions, and continuous refinement throughout the thesis project.
- Focused heavily on modularity, scalability, configurability, and separation of concerns to simplify future extension of additional treatment processes and simulation steps.

WorkoutLogger

- As a team of three developed a full-stack workout logging application using Node.js and Express.js.
- Designed RESTful API endpoints for creating, updating, and retrieving workout sessions.
- Implemented MongoDB for data storage with Mongoose schemas to handle user and workout data.
- Integrated authentication features to enable personalized workout tracking.
- Worked with GitLab CI/CD pipeline and deployed application on Heroku

TECHNICAL SKILLS

Programming:

Java JavaScript C# Typescript

Frameworks & Technologies:

JavaFX Node.js Express.js Vue.js
REST APIs Thymeleaf

Databases:

MongoDB MySQL PostgreSQL
Neo4J GraphDB

Tools & Platforms:

Git Postman Docker Wireshark

RELEVANT EXPERTISE

- Focus areas:** Software Engineering, Databases, Algorithms & Data Structures, Programming Paradigms, Web Development, DevOps practices
- Courses:** Databases, Advanced Databases, Web development, Computer Communication, Testing Debugging and Verification, Computer Security, Object oriented design, Functional programming
- Languages:** English, Swedish, Serbian (fluent in speaking, reading, writing)

EDUCATION

B.Sc. in Computer Science

University of Gothenburg

Gothenburg

Trigon Vale

- As a team of three, designed and developed a 2.5D action-RPG in unity using C#
- Implemented a core game-play loop including exploration, enemy encounters and boss fights
- Integrated RPG systems such as story driven progression, and combat mechanics
- Applied object-oriented-design principles to structure game-play systems such as combat, progression and environment into modular reusable components

EXPERIENCE

Logistics worker

IKEA, Backebol

📅 2019 – Ongoing

📍 Gothenburg, SWEDEN

- Assisting in the daily operations of warehouse logistics, ensuring timely product availability
 - Maintaining inventory accuracy by restocking shelves and organizing goods in storage areas
 - Assisting customers with product pick-ups and handling inquiries about availability
 - Operating forklifts to manage heavy goods and support replenishment processes.
 - Developed strong teamwork and time-management skills in a fast-paced environment.
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Receptionist

Nordic Wellness

📅 2018 – 2019

📍 Gothenburg, SWEDEN

- Welcomed and assisted members to create a positive and professional first impression.
 - Handled membership registrations, payments, and service inquiries with accuracy and efficiency.
 - Managed phone calls, emails and scheduling of different appointments for trainers and trainees.
 - Strengthened organizational and multitasking abilities while working in a customer-focused environment.
 - Built communication skills through daily interaction with clients and staff.
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Welder

Volvo cars

📅 2016 - 2017

📍 Gothenburg, SWEDEN

- Contributed to the demo project for the new Volvo XC60 as part of a collaborative production team.
 - Welded precision components at my station to ensure quality and safety.
 - Gained hands-on experience with technical equipment and compliance with safety standards.
 - Improved attention to detail and consistency by meeting production deadlines and quality requirements.
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